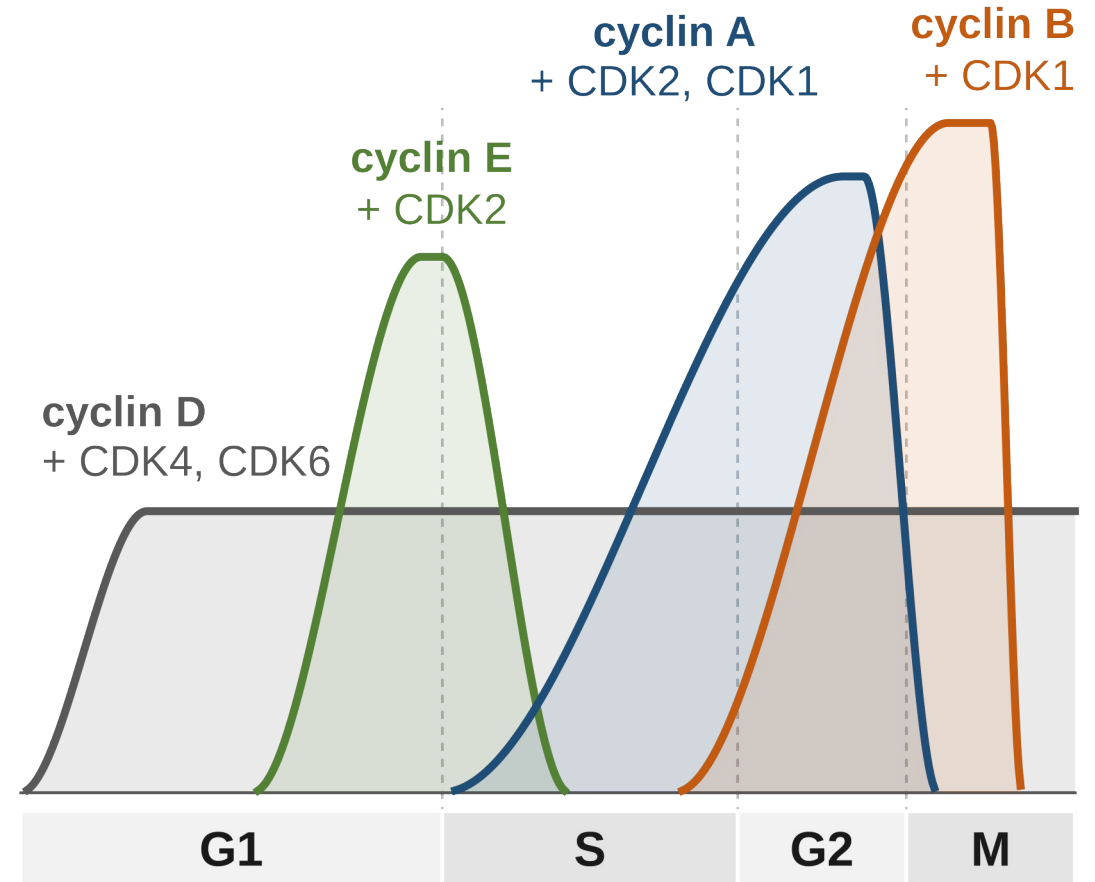


Cyclin-dependent kinases (CDKs)

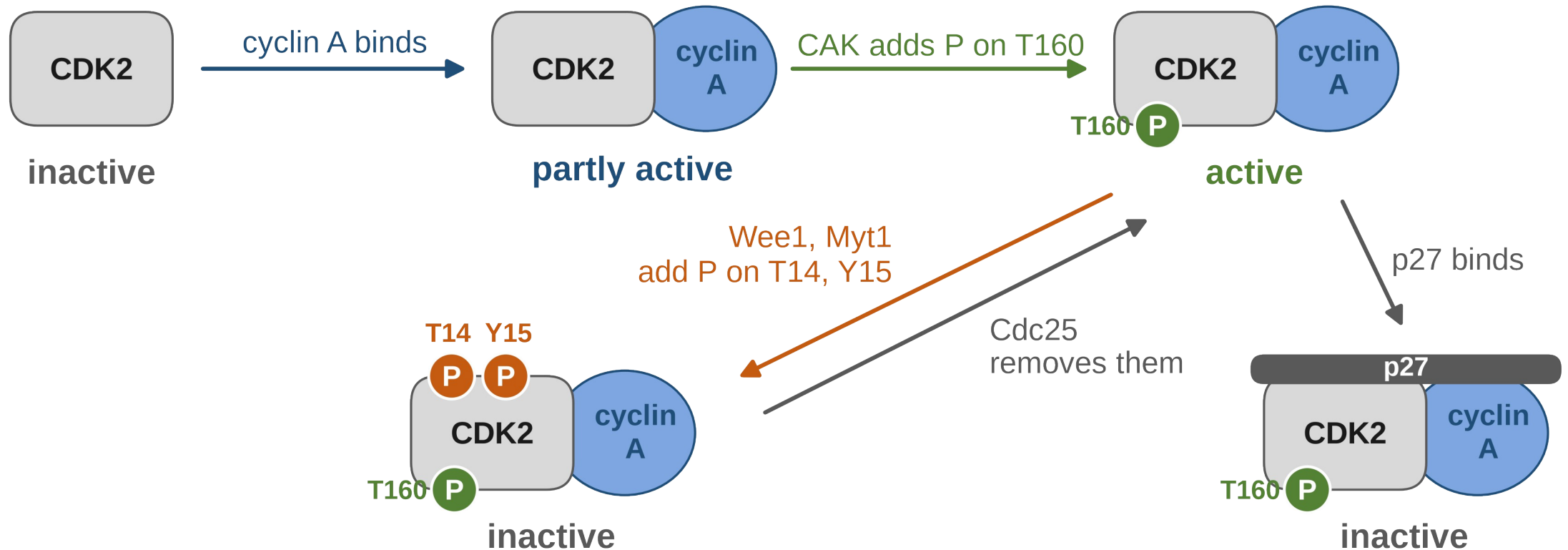
- Protein kinases: move a phosphate from ATP onto Ser/Thr of a target protein
- Active only with a **cyclin** partner
- Cyclins are made and destroyed in each cycle, so CDK activity rises and falls
- CDK7, CDK8, CDK9 and CDK19 control transcription, not the cell cycle

cyclin levels in one cell cycle (schematic)

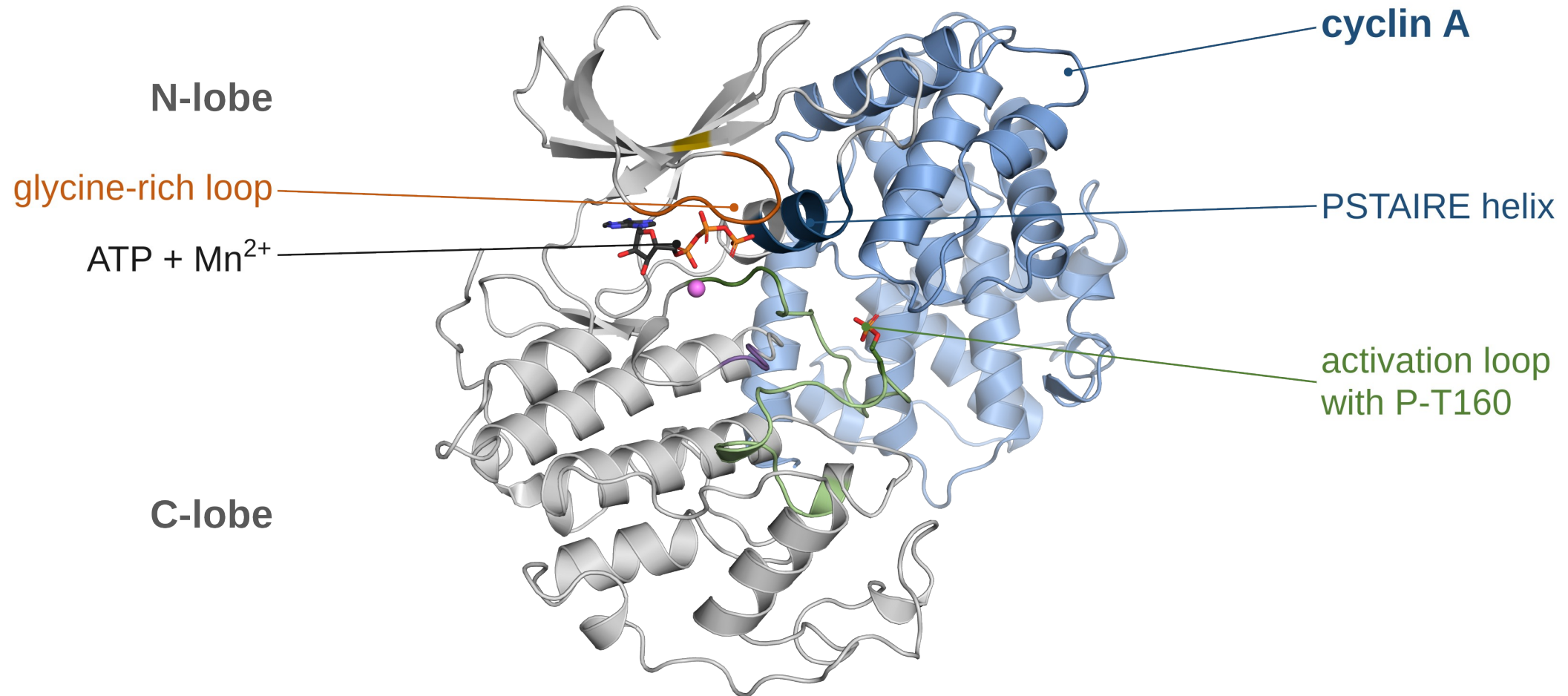


Switching a CDK on and off

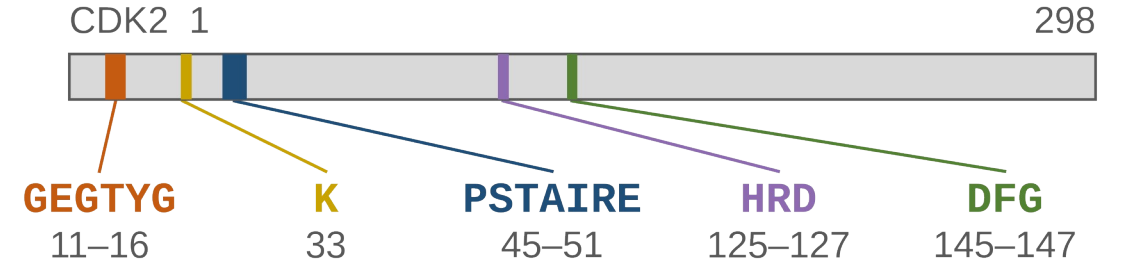
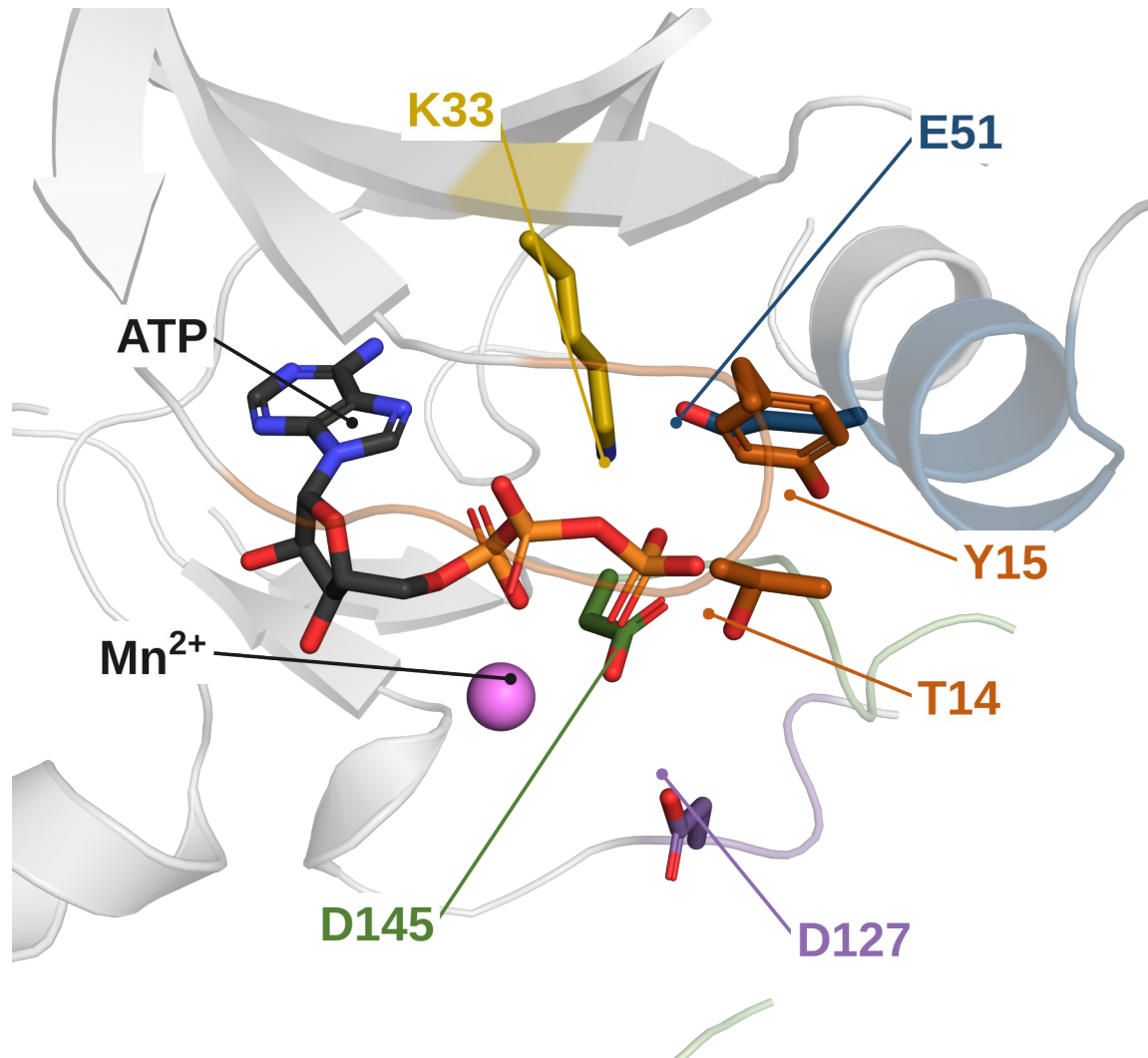
- On: cyclin binding, then a phosphate on T160 added by CAK (CDK7–cyclin H)
- Off: phosphates on T14 and Y15 (Wee1, Myt1) or a bound inhibitor (p21, p27)



CDK2 with cyclin A in 3D



The active site of CDK2



GEGTYG lid over the ATP phosphates;
P on T14 / Y15 blocks it

K K33 holds the ATP phosphates

PSTAIRE binds cyclin; E51 pairs with K33

HRD D127 activates the target Ser/Thr

DFG D145 holds the metal ions of ATP